

Concept Review

Microphone & Speaker

Why Use Speaker and Microphone?

Microphones capture vibrations in air and turn them into electrical and then digital signals, while a speaker performs the reverse. Together, they form the foundation of many real-world systems that listen, process, and respond, enabling applications such as speech recognition, voice communication, environmental monitoring, and acoustic sensing.

Sound Composition and Sine Waves

Sound is a vibration that propagates as a pressure wave through a medium (such as air). Microphones convert these pressure variations into electrical voltage signals, and speakers perform the reverse, converting electrical signals back into acoustic pressure waves.

The simplest form of sound is a pure tone, represented mathematically as a sine wave:

$$x(t) = A \sin(2\pi f t + \phi)$$

Where A is the amplitude (maximum pressure or voltage), f is frequency in Hz (cycles per second), t is time in seconds, and ϕ is phase offset in radians

A single sine wave is a pure tone as it contains only one frequency component. However, most real-world sounds (speech, music, noise) are composite signals consisting of many sine waves of different frequencies and amplitudes.

This is expressed by Fourier's theorem, which states that any periodic signal can be represented as a sum of sine and cosine waves:

$$x(t) = \sum_{k=1}^N A_k \sin(2\pi f_k t + \phi_k)$$

Here, each f_k corresponds to a frequency component in the sound. The fundamental frequency determines the pitch. Harmonics (integer multiples of the fundamental) shape the timbre or tone color. Noise or complex sounds contain a continuous range of frequencies.

Sampling and Digital Audio Fundamentals

When sound is recorded by a microphone, it is converted from a continuous (analog) voltage to discrete digital samples. This process involves sampling and quantization.

Sampling captures the signal amplitude at regular time intervals defined by the sampling rate, f_s :

$$x[n] = x(nT_s)$$

Where $x[n]$ is the discrete form of the signal $x(t)$, and T_s is the sampling period $\frac{1}{f_s}$.

According to the Nyquist-Shannon sampling theorem, the sampling rate must be at least twice the highest frequency component in the signal to avoid aliasing. If this condition is not met, aliasing occurs — higher frequency components fold back into lower frequencies, distorting the signal.

Typical sample rate for telephone is 8 kHz, while CD-quality audio has a sample rate of 44.1 kHz. Nowadays, common professional and embedded applications are 48 kHz.

Each sampled analog value is approximated by the nearest available digital level. The number of possible levels, also called the resolution, depends on bit depth N_b , and is equal to 2^{N_b} . A higher bit depth improves amplitude resolution and reduces quantization noise. For example, the classic 8-bit sounds have coarse resolution (256 levels), while current standard audio streaming services use 16-bit sounds (65,536 levels).

In practice, audio data is processed in blocks (or frames) of samples, rather than one sample at a time. For a block size of N samples and a sample rate f_s , the duration of each block is:

$$T_{block} = \frac{N}{f_s}$$

Smaller block sizes lead to faster updates and lower latency, but higher CPU load, while larger block sizes create smoother visualization and better frequency resolution in FFT, but slower updates. These trade-offs are important for real-time applications like sound visualization and event detection.

Fast Fourier Transform (FFT)

While the time-domain representation of a signal $x(t)$ or $x[n]$ shows how amplitude varies over time, it often hides the *frequency content* that defines the sound's pitch, tone, and harmonic structure. Many important sound features are better understood by transforming the signal into the frequency domain, which shows how much of each frequency is present, as shown in Figure 1.

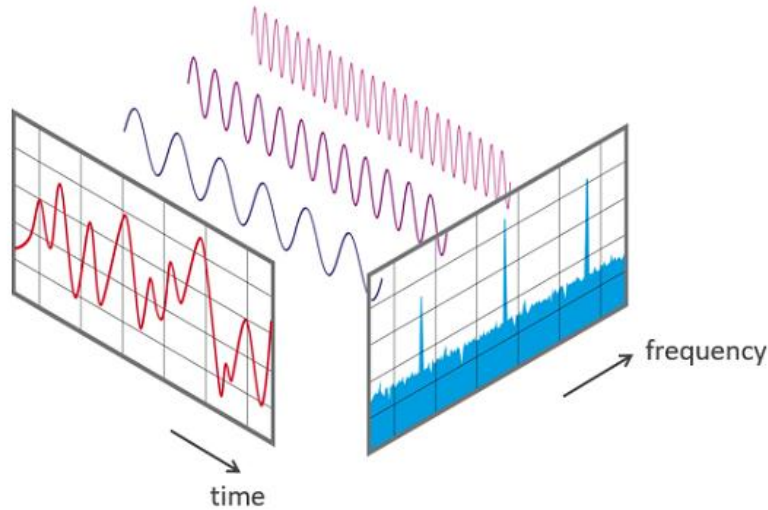


Figure 1. A complex signal composed of sinusoids in time and frequency domain.

For a continuous-time signal $x(t)$, Fourier Transform decomposes it into its frequency components. The transform projects the signal onto sinusoidal basis functions to determine how much of each frequency is present. In digital systems, we cannot handle continuous signals, so we use the **Discrete Fourier Transform (DFT)** to analyze sampled data $x[n]$, $n = 0, 1, \dots, N - 1$:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-\frac{i2\pi kn}{N}}, \quad k = 0, 1, \dots, N - 1$$

Where $x[n]$ is the time-domain signal, $X[k]$ is the DFT coefficient at frequency bin $f_k = k\Delta f$, with Δf being the frequency resolution $\Delta f = f_s/N$.

For example, at a sampling rate of $f_s = 48,000$ Hz and FFT block size $N = 1024$:

$$\Delta f = \frac{44100}{1024} \approx 43.1\text{Hz}$$

Since $X[k]$ is complex, it contains both magnitude and phase information:

$$|X[k]| = \sqrt{(\text{Re}\{X[k]\})^2 + (\text{Im}\{X[k]\})^2}$$

$$\angle X[k] = \tan^{-1}\left(\frac{\text{Im}\{X[k]\}}{\text{Re}\{X[k]\}}\right)$$

In practice, the magnitude spectrum shows how strong each frequency component is. The phase spectrum shows timing and alignment information, but it's often omitted for basic audio analysis.

The Fast Fourier Transform (FFT) is an efficient algorithm that computes the DFT. In practice, the FFT produces the same result as the DFT, but far more efficiently. In modern software environments, you rarely implement FFT manually because optimized libraries are used instead. In Python, several libraries contain an FFT function, and common options include **numpy**, **scipy**, and **torch**.

Selecting a Microphone and Speaker

Microphones and speakers are used across a variety of applications, such as recording, streaming, or as part of consumer devices. Specific requirements will depend factors such as on the size and ambient noise of the environment, and the content that the device is intended to capture or reproduce.

Total harmonic distortion (THD): Harmonic distortion refers to the unwanted components at frequencies, called harmonic frequencies, that are integer multiples of the original signal's fundamental frequency. The total harmonic distortion (THD), expressed as a percentage, refers to the ratio of the summed power of harmonic frequencies to the power of the fundamental frequency. For microphones, distortion occurs when sound is added

Maximum sound pressure level: The sound pressure level (SPL) is a refers to the pressure deviation caused by a sound wave, measured in decibels (dB). Subjectively, this is perceived as the "loudness" of a sound. Microphone and speaker datasheets will list the maximum SPL that the device can capture or produce before the distortion (THD) exceeds a certain threshold, which is usually 0.5% or 1%.

Frequency response: This is the sensitivity of the microphone's response to sounds across a range of pitches (frequencies). For a speaker, this refers to how accurately it can reproduce sounds across a range of pitches (frequencies). A flat line frequency response plot represents a more balanced sound, but certain applications may benefit from a microphone or speaker that attenuates or amplifies certain frequencies more than others.

Sensitivity: This is the measure of how microphones convert an amount of sound pressure to a signal voltage, usually measured in millivolts per Pascal (mV/Pa) or decibels (dB) relative to 1V/Pa. Microphones with a higher sensitivity, indicated by a higher number, may be better suited for smaller rooms since it can pick up quieter sounds with more detail, but it may also pick up more background noise.

For speakers, the sensitivity refers to the sound output of the speaker, expressed as number of dB using 1 W of power at a distance of 1 meter. A higher sensitivity means that the speaker requires less power to produce a certain sound pressure level compared to a lower sensitivity speaker.

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